List Manipulation — Teaching Notes

Class XI Computer Science with Python — Chapter 7 (Sumita Arora)

Learning Objectives

By the end of this chapter, students should be able to:

- Create lists in different ways and explain why lists are **mutable**
- Access, slice, and traverse list elements using indexing (forward and backward)
- Perform list operations: joining (+), repetition (*), and slicing
- Use built-in list methods to add, update, delete, search, sort, and count elements
- Explain the difference between an **alias** and a **true copy** of a list
- Write short programs that process lists (search, min/max, mean, frequency count)

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1. What is a List? (7.1 Introduction)

A **list** is Python's built-in data type for storing an ordered sequence of values — of **any** type, and even of **mixed** types, in the same list.

```
[]                          # empty list
[1, 2, 3]                   # list of integers
[1, 2.5, 3.7, 9]             # mixed int/float
['a', 'b', 'c']              # list of characters
['a', 1, 'b', 3.5, 'zero']   # mixed types
```

Key idea to emphasise: Lists are **mutable** — you can change an element in place without creating a new list object. This is the single biggest conceptual difference from strings and tuples (which are immutable), and it drives a lot of "gotcha" behaviour later (aliasing, in-place methods returning `None`, etc.).

> **Teaching tip:** Draw two boxes on the board — "Mutable: list, dict" and "Immutable: int, float, str, tuple" — and keep referring back to it all chapter.

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2. Creating and Accessing Lists (7.2)

2.1 Creating Lists

Use square brackets with comma-separated values — this is called a **list display construct**:

```
L = []                 # empty list, same as L = list()
L = [value, value, ...]
```

Special cases to walk through:

Case	Example	Note
Empty list	`L = []` or `L = list()`	Truth value is `False`, like `0` or `''`
Long list across lines	`sqrs = [0, 1, 4, 9,\n 16, 25, 36]`	Python only needs matching `[ ]`, line breaks are fine
Nested list	`L1 = [3, 4, [5, 6], 7]`	`L1` has **4** elements — `[5, 6]` counts as *one* element. `len(L1)` is 4, and `L1[2]` is the list `[5, 6]`

Creating a list from an existing sequence — `list(<sequence>)`:

```
>>> list('hello')
['h', 'e', 'l', 'l', 'o']      # from a string

>>> t = ('w', 'e', 'r', 't', 'y')
>>> list(t)
['w', 'e', 'r', 't', 'y']      # from a tuple
```

Reading a list from the keyboard — two common approaches:

```
# 1) list(input(...)) — splits a string into individual characters
l1 = list(input('Enter list elements: '))
# Input: 234567  →  Output: ['2','3','4','5','6','7']  (all elements are
strings!)

# 2) eval(input(...)) — the more useful approach for real lists/tuples
list1 = eval(input("Enter list to be added: "))
# Input: [67, 78, 46, 23]  →  Output: [67, 78, 46, 23]  (a genuine list of
ints)
```

About `eval()`: `eval()` evaluates a string as a Python expression and returns the *actual* result with its intended type.

```
>>> eval('5 + 8')
13
>>> var1 = eval(input("Enter Value: "))      # Input: 15 + 3  →  18 <class
'int'>
```

Use `[ ]` inside the input to read a list, and `( )` to read a tuple via `eval(input())`.

> ⚠️ **Caveat worth mentioning to students:** `eval()` is powerful but risky/unpredictable (it executes arbitrary code), and the textbook itself notes it is "always discouraged" in production code — use it mainly for quick classroom/learning exercises.

2.2 Accessing Lists

Lists behave like strings for indexing — **forward indexing** starts at 0, **backward indexing** starts at -1.

```
list1 =  ['a',  'e',  'i',  'o',  'u']
Index:     0     1     2     3     4      (forward)
Index:    -5    -4    -3    -2    -1      (backward)
```

```
>>> vowels = ['a', 'e', 'i', 'o', 'u']
>>> vowels[0]
'a'
>>> vowels[-1]
'u'
>>> vowels[5]
IndexError: list index out of range
```

Mutability in action — unlike a string, you *can* assign to an index:

```
>>> vowels[0] = 'A'
>>> vowels
['A', 'e', 'i', 'o', 'u']      # changed IN PLACE — no new list object
created
```

Traversing a list with a `for` loop:

```
L = ['P', 'y', 't', 'h', 'o', 'n']
for a in L:
    print(a)
```

or by index, using `range(len(L))` when you need the position too:

```
L = ['q', 'w', 'e', 'r', 't', 'y']
length = len(L)
for a in range(length):
    print("At indexes", a, "and", (a - length), "element:", L[a])
```

```
At indexes 0 and -6 element : q
At indexes 1 and -5 element : w
...
```

Comparing lists: Python compares lists **element by element, lexicographically**. Equality (`==`, `!=`) needs matching element types; ordering (`<`, `>`, etc.) requires each pair of corresponding elements to be of *comparable* types, otherwise Python raises `TypeError`.

```
>>> [1, 2, 3] == [1, 2, 3]
True
>>> [1, 2, 8, 9] < [1, 2, 9, 1]     # compares 3rd elements: 8 < 9
True
>>> [2, 3] < ['2', '3']
TypeError: '<' not supported between instances of 'int' and 'str'
```

> **Teaching tip:** The comparison stops at the **first differing pair** of elements — it doesn't matter how "big" later elements are. Work through Table 7.1 style examples on the board.

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3. List Operations (7.3)

3.1 Joining Lists — the `+` operator

```
>>> lst1 = [1, 3, 5]
>>> lst2 = [6, 7, 8]
>>> lst1 + lst2
[1, 3, 5, 6, 7, 8]
```

`+` requires both operands to be lists. `list + number`, `list + string` etc. all raise `TypeError`.

3.2 Repeating/Replicating Lists — the `*` operator

```
>>> [1, 3, 5] * 3
[1, 3, 5, 1, 3, 5, 1, 3, 5]
```

`*` needs a list and an **integer**.

> **Memory aid:** `+` wants (list, list). `*` wants (list, int).

3.3 Slicing Lists

```
seq = L[start : stop]          # elements at start, start+1, ..., stop-1
seq = L[start : stop : step]   # same, but skipping (step-1) elements each
time
```

```
>>> lst = [10, 12, 14, 20, 22, 24, 30, 32, 34]
>>> lst[3:-3]
[20, 22, 24]
>>> lst[0:10:2]        # every 2nd element
[10, 14, 22, 30, 34]
>>> lst[::3]            # every 3rd element, full list
[10, 20, 30]
>>> lst[::-1]           # step -1 → reverses the list
[34, 32, 30, 24, 22, 20, 14, 12, 10]
```

Important behaviour to highlight: unlike plain indexing, **out-of-range slice bounds never raise an error** — Python just clips to whatever exists (or returns `[]` if nothing falls in range):

```
>>> lst[3:30]        # upper bound way beyond length — no error
[20, 22, 24, 30, 32, 34]
>>> lst[10:20]        # both bounds out of range
[]
```

A slice is itself a list (not a copy-only view) — you can modify it independently:

```
>>> seq = lst[3:-3]
>>> seq[1] = 28
>>> seq
[20, 28, 24]          # only affects 'seq', not 'lst'
```

Using slices to modify a list — assign a sequence to a slice to overwrite a *range* of elements, possibly with a different number of elements:

```
>>> L = ["one", "two", "THREE"]
>>> L[0:2] = [0, 1]
>>> L
[0, 1, 'THREE']

>>> L = ["one", "two", "THREE"]
>>> L[0:2] = "a"                 # a string is also a sequence!
>>> L
['a', 'THREE']
```

The right-hand side **must be an iterable/sequence** — `L[2:] = 345` raises `TypeError: can only assign an iterable`.

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4. Working With Lists (7.4)

Task	Syntax	Notes
Append one item	`L.append(item)`	Adds **one** item to the end
Update an item	`L[index] = new_value`	Direct index assignment
Delete by index/slice	`del L[index]` / `del L[start:stop]`	`del L` deletes the whole list object
Delete & retrieve	`L.pop(index)`	Removes and **returns** the item; no index → removes last item

```
>>> lst1 = [10, 12, 14]
>>> lst1.append(16)
>>> lst1
[10, 12, 14, 16]

>>> lst1[2] = 24          # update
>>> lst1
[10, 12, 24, 16]

>>> del lst1[1]           # delete by index
>>> item = lst1.pop()     # remove & store the last element
```

Aliasing vs True Copy — a classic exam/conceptual trap

```
>>> a = [1, 2, 3]
>>> b = a                 # b is just ANOTHER NAME for the same list (alias)
>>> a[1] = 5
>>> b
[1, 5, 3]                 # b changed too!
```

To make an **independent** copy:

```
>>> a = [1, 2, 3]
>>> b = list(a)           # or a[:] or a.copy()
>>> a[1] = 5
>>> b
[1, 2, 3]                 # b is unaffected now
```

> **Teaching tip:** Draw the two-arrows-pointing-to-one-box diagram for aliasing, then the two-separate-boxes diagram for a true copy. This single picture clears up 90% of student confusion here.

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5. List Functions and Methods (7.5)

> Syntax convention below: `List` means any list object/literal; replace with an actual list name.

#	Method	Syntax	Arguments / Returns	What it does
1	**index**	`L.index(item)`	1 arg, returns index (int)	Index of **first** match; `ValueError` if absent
2	**append**	`L.append(item)`	1 arg, returns `None`	Adds **one** element (even a list!) to the end
3	**extend**	`L.extend(list2)`	1 arg (must be a list), returns `None`	Adds **each element** of `list2` to the end
4	**insert**	`L.insert(pos, item)`	2 args, returns `None`	Inserts `item` *before* index `pos`

#	Method	Syntax	Arguments / Returns	What it does
5	**pop**	`L.pop(index)`	1 optional arg, **returns** removed item	Removes & returns item at `index` (default: last)
6	**remove**	`L.remove(value)`	1 arg, returns `None`	Removes **first occurrence** of `value`; `ValueError` if absent
7	**clear**	`L.clear()`	no args, returns `None`	Empties the list — list object still exists
8	**count**	`L.count(item)`	1 arg, returns count (int)	Number of occurrences of `item`; `0` if absent
9	**reverse**	`L.reverse()`	no args, returns `None`	Reverses the list **in place**
10	**sort**	`L.sort()` / `L.sort(reverse=True)`	optional arg, returns `None`	Sorts **in place**, ascending by default

append() vs extend() — the method students mix up most

```
>>> t1 = ['a', 'b', 'c']
>>> t2 = ['d', 'e']
>>> t1.extend(t2)
>>> t1
['a', 'b', 'c', 'd', 'e']      # each element of t2 added individually

>>> t1 = [1, 3, 5]
>>> t1.append([12, 14])
>>> t1
[1, 3, 5, [12, 14]]            # the WHOLE list added as ONE element (nested
list!)
>>> len(t1)
4
```

	`append(x)`	`extend(list2)`
Adds	exactly one object (even if that object is itself a list)	every element of `list2`, one by one

	`append(x)`	`extend(list2)`
Length increases by	1	`len(list2)`
Argument type	anything, single value	must be an iterable/list

insert() — three special behaviours worth drilling

```
L.insert(0, x)        # inserts x at the very front
L.insert(len(L), x)   # same effect as L.append(x)
L.insert(-9, x)        # index beyond valid negative range → prepends x to
the front
```

pop() vs remove() — the other classic pair

	`pop(index)`	`remove(value)`
Argument is	a **position/index** (optional)	a **value**
Returns	the removed item	nothing (`None`)
Error if	list is empty	value not found

```
>>> t1 = ['k', 'a', 'e', 'i', 'p', 'q', 'u']
>>> t1.pop(0)            # removes & returns the item AT index 0
'k'
>>> t1.remove('p')        # removes the first occurrence of VALUE 'p'
```

sort() and reverse() both mutate in place and return None

```
>>> t1 = ['e', 'i', 'q', 'a', 'q', 'p']
>>> t1.sort()
>>> t1
['a', 'e', 'i', 'p', 'q', 'q']

>>> t1.sort(reverse=True)     # descending order
```

> ⚠️ `sort()` cannot order a list containing complex numbers — Python has no ordering relation defined for them.

> **Common student bug:** writing `newlist = L.sort()` or `newlist = L.append(x)` and then being confused that `newlist` is `None`. These methods modify `L` in place and return nothing — always emphasise this.

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6. Worked Example Programs

These are good board/lab exercises — ask students to trace them by hand first, then run.

(a) Find the minimum element and its index

```
lst = eval(input("Enter list: "))
length = len(lst)
min_ele = lst[0]
min_index = 0
for i in range(1, length):
    if lst[i] < min_ele:
        min_ele = lst[i]
        min_index = i
print("Minimum element is:", min_ele, "at index", min_index)
```

(b) Calculate the mean of a list of numbers

```
lst = eval(input("Enter list: "))
length = len(lst)
total = 0
for i in range(length):
    total += lst[i]
mean = total / length
print("Mean:", mean)
```

(c) Linear search for an element

```
lst = eval(input("Enter list: "))
element = int(input("Enter element to search: "))
for i in range(len(lst)):
    if element == lst[i]:
        print(element, "found at index", i)
        break
else:                                # for-loop's else: runs only if NO
break happened
    print(element, "not found in list")
```

(d) Count the frequency of a given element

```
lst = eval(input("Enter list: "))
element = int(input("Enter element: "))
count = 0
for i in range(len(lst)):
    if element == lst[i]:
        count += 1
print(element, "occurs", count, "times")
```

(e) Separate unique vs duplicate elements (good extension/challenge task)

```
lst = eval(input("Enter list: "))
uniq = []
dupl = []
for item in lst:
    if lst.count(item) == 1:
        uniq.append(item)
    elif item not in dupl:
        dupl.append(item)
print("Unique elements:", uniq)
print("Duplicate elements:", dupl)
```

(This is a simplified, teaching-friendly version of the book's while-loop based program 7.7 — using `count()` makes the logic easier for first-time learners to follow; introduce the book's longer version as a 'how would you do this without count()?' extension.)

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7. Common Misconceptions to Address in Class

1. **"append() and extend() do the same thing."** — They don't; `append` nests, `extend` flattens in the elements.
2. **"b = a makes a copy of list a."** — It doesn't; it's an alias. Use `b = list(a)`, `b = a.copy()`, or `b = a[:]` for a true copy.
3. **"Methods like sort(), reverse(), append() return the new list."** — They return `None`; the change happens in place.
4. **"List slices raise IndexError like direct indexing."** — They don't; out-of-range slice bounds are silently clipped.
5. **"Strings and lists are accessed/sliced completely differently."** — They're very similar (same indexing/slicing rules); the real difference is **mutability**.
6. **"pop() and remove() are interchangeable."** — `pop` works on position, `remove` works on value.

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8. Quick Revision Checklist (for students)

- [ ] Define a list; explain mutability vs immutability
- [ ] Create lists: literal, empty, nested, from `list(sequence)`, via `eval(input())`
- [ ] Access elements with positive/negative indexing
- [ ] Traverse a list with `for item in L` and with `for i in range(len(L))`
- [ ] Slice a list: `L[start:stop]`, `L[start:stop:step]`, `L[::-1]`
- [ ] Join lists with `+`, replicate with `*`
- [ ] Modify lists via index assignment and slice assignment
- [ ] Use `append, extend, insert, pop, remove, clear, index, count, sort, reverse`
- [ ] Distinguish alias (`b = a`) from true copy (`b = list(a)`)
- [ ] Write simple list-processing programs: search, min/max, mean, frequency count

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9. Sample Practice Questions (adapted from Checkpoint 7.1)

1. Why are lists called mutable types? What are their immutable counterparts?
2. What are the different ways of creating a list?
3. If `a = [5, 4, 3, 2, 1, 0]`, evaluate: `a[0]`, `a[-1]`, `a[a[0]]`, `a[a[-1]]`
4. What is the difference between `append()` and `insert()`?
5. What is the difference between `pop()` and `remove()`?
6. What is the difference between `append()` and `extend()`?
7. How does `sort()` work? Give an example with `reverse=True`.
8. What does `list.clear()` do, and how is it different from `del list`?